



Apple blocks Fortnite updates

Apple terminated Epic Games' App Store developer account, blocking all future Fortnite updates. <http://dgit.in/sep20-11>



Poco X3 will have a 64MP lens

A Poco company executive confirmed on Twitter that the Poco X3 will feature a 64MP primary lens. <http://dgit.in/sep20-12>

MSI MEG Z490 ACE

A good mix of gaming and overclocking features

Most motherboards based on the Z490 platform have a similar feature set as their older Z390 counterparts given that the platform hasn't evolved much and the fact that the Comet Lake processors are still based on the Skylake architecture with a few additional cores in the top SKUs. The MSI MEG Z490 ACE in particular sits right between the absolute flagship MEG Godlike and the more sobre MPG Gaming Carbon Wi-Fi. We've spotted the board selling for INR 43,999 which tells us that MSI is attempting to strike a balance between packing in more features while reeling in the cost.

Board Layout

The MSI MEG Z490 ACE has a very busy layout with a lot of components and ancillary components being packed onto the ATX form-factor.



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Price
₹43,999

the rear IO shield. MSI does not use a dense fin stack on the MEG Z490 ACE which is a bit disappointing given that this is a premium board. The Z490 GODLIKE has traded the blocks for fin stacks and it would have been better if the Z490 ACE also had a fin stack.

CPU Area and VRM

For the VRM, the MEG Z490 ACE uses an Intersil ISL69269 Digital PWM controller which then handles all the power stages. The ISL69269 can be implemented in a 8+2 or 8+1+1 configuration. In this case, MSI uses Intersil ISL99390 90A power stages and we can see 16 of these babies along the top and left edge of the CPU socket. So MSI does not use a doubler and has instead opted to mirror 8 power phases, hence, the use of 16 ISL99390. We recently saw a shift across all motherboard manufacturers as they moved away from using doublers on the more premium boards to reduce ripple, lower temps and build a more sturdy VRM.

Sitting on top of the VRM are massive chunky metal heatsinks that are attached to one another using a heat pipe. The block on the left even has a little fan blowing air onto it from

Power and RAM

The MSI MEG Z490 ACE makes use of two 8-PIN EPS connectors which then feed into the PWM controller. This allows for the board to distribute the load across multiple connectors and helps control heat generation at the EPS power connectors. Intel's top Comet Lake processors have a base TDP of 125W which doesn't tell you how much power it consumes but it gives you a fair idea of what it might under load. We've seen peak power loads of 290+ watts on the Intel Core i9-10900K which is quite a lot for a single 8-pin power connector to handle over an extended period of time.

The board dedicates 2 power phases for the memory which is sufficient for all RAM configurations. Overclockers tend to use a 2xDIMM configuration anyway.

Right next to the RAM slots we also see the voltage test pads as well

as POST status LEDs for the key components. Aside from that MSI also has the alphanumeric EZ debug LED indicator which is present on the bottom of the board.

Expansion slots

The MSI MEG Z490 ACE has 3 x16 PCIe slots, 2 x1 PCIe slots and 3 NVMe slots. The first two of the PCIe slots are from the CPU and the third one comes from the PCH. These can be configured in either a 16/0/4 arrangement or in an 8/8/4 arrangement. The x1 PCIe slots come from the PCH. So you can implement a 2-way NVIDIA SLI or a 3-way AMD CrossFire multi-GPU setup if you need the ridiculous amount of horsepower a multi-GPU setup provides.

Audio

On the Z490 ACE, we see the audio circuitry is powered by a Realtek ALC1220 CODEC coupled with the ESS Sabre ES9018 DAC. Plus the Nahimic software solution is always a good addition.

Verdict

The MSI MEG Z490 ACE is quite a pricey board for the Indian markets but all features considered, this ends up being a top-end motherboard which is at par with all the other offerings from competing manufacturers. The board does not have any features for folks implementing custom loops but for someone wanting to run the latest Intel Comet Lake processor at its peak capabilities and go further with overclocking, the Z490 ACE is a great board.

—Mithun Mohandas

SPECIFICATIONS

CHIPSET: Intel Z490 | MEMORY SUPPORT: 4x5000(OC) MHz upto 128GB | LAN: Realtek 8125B 2.5G and Intel I219V Gigabit Ethernet | AUDIO: ALC 1220 + ESS SABRE9018Q2C | SATA: 6x SATA III | EXPANSION: 3x PCIe x16(x16+x0+x4) | DIMENSIONS (LXWXD): 305 mm x 244 mm | WARRANTY: 3 years.

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